



## COMPANY PROFILE

# **AXIS WELL DELIVERY SYSTEMS LTD**

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## **INTEGRATED WELL DELIVERY COMPANY PROFILE**

Engineered Right, Creating Value

Nigeria • West Africa • Beyond  
[www.axiswelldelivery.com](http://www.axiswelldelivery.com)

## About Us

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**Axis Well Delivery Systems Ltd** is a premier indigenous oilfield services company delivering integrated, turnkey solutions to Nigeria's upstream petroleum industry. Headquartered in Nigeria, we specialize in bridging the gap between **complex subsurface challenges and efficient, reliable production outcomes**.

Our capabilities span the full well lifecycle — from drilling engineering and well construction to completions, intervention, and production optimization. Through our integrated delivery model, we enable operators to streamline vendor interfaces, reduce operational risk, and achieve cost-efficient well execution.

Axis is uniquely positioned to support the evolving Nigerian upstream landscape, with a strong focus on brownfield revitalization and marginal field development. By combining deep local operational insight with global engineering standards, we consistently deliver fit-for-purpose solutions that unlock reservoir value and accelerate time to first oil.

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*"We do not just deliver wells — we deliver producing assets, operational confidence, and long-term value creation."*

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## Our Vision

To be the foremost indigenous partner of choice in Africa, redefining excellence in integrated well delivery and sustainable asset optimization.

## Our Mission

To deliver superior value to upstream operators by providing innovative, end-to-end well engineering and production solutions. We are committed to maximizing the potential of marginal and brownfield assets through technical precision, operational safety, and a deep-rooted understanding of the African energy landscape.

## Our Core Value Pillars

|          |                                             |                                                                                                                                                                                                |
|----------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | <b>Local Insight,<br/>Global Standards</b>  | We combine indigenous operational intelligence with internationally benchmarked engineering practices to deliver solutions tailored to the Niger Delta and similar environments.               |
| <b>2</b> | <b>Operational Integrity</b>                | We operate under a strict Goal Zero philosophy, embedding HSSE excellence, risk management, and process discipline into every phase of well delivery to ensure safe, incident-free operations. |
| <b>3</b> | <b>Asset Maximization</b>                   | We focus on extending asset life and improving recovery efficiency, applying tailored completion strategies, production optimization, and intervention planning to maximize economic returns.  |
| <b>4</b> | <b>Integrated Execution Excellence</b>      | Our integrated project delivery model reduces interface risk, enhances coordination, and ensures seamless execution from planning through production.                                          |
| <b>5</b> | <b>Cost Efficiency &amp; Value Delivery</b> | We are committed to delivering wells on time and within budget, leveraging lean organizational structures, optimized workflows, and proactive NPT management.                                  |
| <b>6</b> | <b>Our Promise</b>                          | To deliver wells safely, efficiently, and profitably — transforming subsurface potential into sustainable production.                                                                          |

## Our Services

We provide comprehensive well lifecycle solutions, combining technical excellence with operational efficiency to deliver measurable results for our clients across Nigeria's upstream sector.

### Integrated Well Delivery (IPM)



Our Integrated Project Management offering covers the complete well delivery cycle, providing single-point accountability from planning through production.

We coordinate all vendors, track costs against AFE, and manage risk proactively.

- Full lifecycle well delivery (planning → execution → production)
- Vendor coordination & contract management
- Cost control & AFE tracking with real-time reporting
- Risk management & contingency planning

### Drilling Engineering

Our drilling engineering team designs wells to maximize reservoir access while minimizing operational risk. We leverage advanced torque & drag modeling, hydraulics optimization, and structured DWOP facilitation to ensure every well is drilled safely and efficiently.

- Well design: vertical, deviated, and horizontal trajectories
- Torque & drag modeling for complex well paths
- Hydraulics optimization for efficient hole cleaning
- Well programs & DWOP facilitation with all stakeholders

## Completions & Sand Control

Sand management is critical in Niger Delta formations. Our completions team designs and executes gravel pack, frac pack, and standalone screen installations tailored to formation characteristics. We also engineer artificial lift systems and manage well integrity throughout the asset lifecycle.

- Gravel pack / frac pack design optimized for formation sand properties
- Tubing & completion design for single and dual-string configurations
- Artificial lift selection and design: ESP, PCP, and gas lift systems
- Well integrity & barrier management throughout asset life

## Well Intervention

When wells underperform or require remediation, our intervention team provides rapid, cost-effective solutions using slickline, electric line, and coiled tubing technologies. We diagnose well problems accurately and execute targeted interventions to restore or enhance production.

- Slickline / E-line operations for mechanical and data acquisition tasks
- Coiled tubing services for wellbore cleanup, stimulation, and nitrogen kicks
- Well diagnostics & remediation using PLT, production logging, and downhole cameras

## Production Optimization

We help operators extract maximum value from their wells through continuous performance monitoring, artificial lift optimization, and surface facility integration. Our data-driven approach identifies production constraints and implements solutions that deliver measurable uplift.

- Well performance analysis using nodal analysis and IPR/VLP modeling
- Artificial lift optimization to maximize run life and production rates
- Surface facility integration ensuring wellhead-to-pipeline efficiency

# Our Turnkey Delivery Model

Our four-phase framework eliminates interface risks and accelerates the First Oil timeline. Each phase builds upon the previous, creating a seamless progression from subsurface assessment to sustained production.

| Phase           | Core Objective        | Timeline Focus        | Key Deliverables                                                                                    |
|-----------------|-----------------------|-----------------------|-----------------------------------------------------------------------------------------------------|
| I. Engineering  | De-risking the Asset  | Pre-spud              | Well Trajectory & Casing Design, Geomechanics, AFE Optimization, DWOP Facilitation                  |
| II. Execution   | Operational Precision | Drilling & Completion | Rig/Service Integration, Real-time NPT Management, Dual-String/Gravel Pack Completions              |
| III. Activation | Production Readiness  | Post-completion       | Engineered Unloading (N <sub>2</sub> Kick-off), Artificial Lift (ESP/PCP) Readiness, Surface Tie-in |
| IV. Lifecycle   | Asset Optimization    | Ongoing               | Continuous Production Monitoring, Sand Control Performance, Intervention Planning                   |

## Our Execution Intelligence

 **Operational Mastery:** Proven leadership in the complex Niger Delta terrain across land, swamp, and offshore.

 **Integrated Control:** Single-point accountability for third-party vendors (SLB, Weatherford, etc.).

 **Regulatory & Community Synergy:** Deep alignment with NUPRC standards and local security frameworks.

 **Digital Integration:** Future-ready operations using Softedge Analytics Borestack™ for data-driven decision making.

“We don’t just supervise wells. We design, execute, and deliver producing assets.”

## Why Choose Axis

We are your trusted partner for integrated well delivery, from concept to production.

|                              |                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>20+ Years Experience</b>  | Proven leadership in Niger Delta operations across land, swamp, and offshore environments with deep understanding of the geological and operational challenges.           |
| <b>Proven Track Record</b>   | Successful execution across Emohua Field (Land Asset), Kuri Field (Swamp Asset), and Ede Field (Offshore Asset) — demonstrating versatility across all Nigerian terrains. |
| <b>Regulatory Excellence</b> | Strong interface with NUPRC and host communities. Deep alignment with local security frameworks including Tantita and community engagement models.                        |
| <b>Turnkey for Marginals</b> | Complete well delivery capability for marginal field operators — from award to first oil, we handle the entire wells scope as your integrated technical partner.          |
| <b>Digital Integration</b>   | Integration-ready with digital oilfield platforms (Softedge Analytics - Borestack™) for real-time data-driven decision making and operational performance tracking.       |



## HSE Commitment — Goal Zero

Axis Well Delivery Systems Ltd operates under a strict **Goal Zero philosophy**, embedding HSSE excellence into every phase of our operations:

**Zero Harm to Personnel**

**Zero Environmental  
Incidents**

**Zero Asset Damage**

### Implementation Framework

We implement the following safety and risk management processes across all our operations to ensure consistent, incident-free well delivery:

 HAZID / HAZOP — Comprehensive hazard identification and operability studies before every operation

 JSA / Risk Assessment — Job safety analysis with documented risk mitigation for each task

 SIMOPS Planning — Structured simultaneous operations planning to manage multi-activity interfaces

 Emergency Response Systems — Tested and drilled response plans with community integration

# Flagship Case Studies

## Case Study 1: Emohua Field (PPL 201)

### Land Asset — Dual-String Completion

**Project Objective:** Maximize reservoir recovery and data acquisition through a dual-string completion strategy across multiple zones in a legacy field environment.

### Engineering Scope Delivered

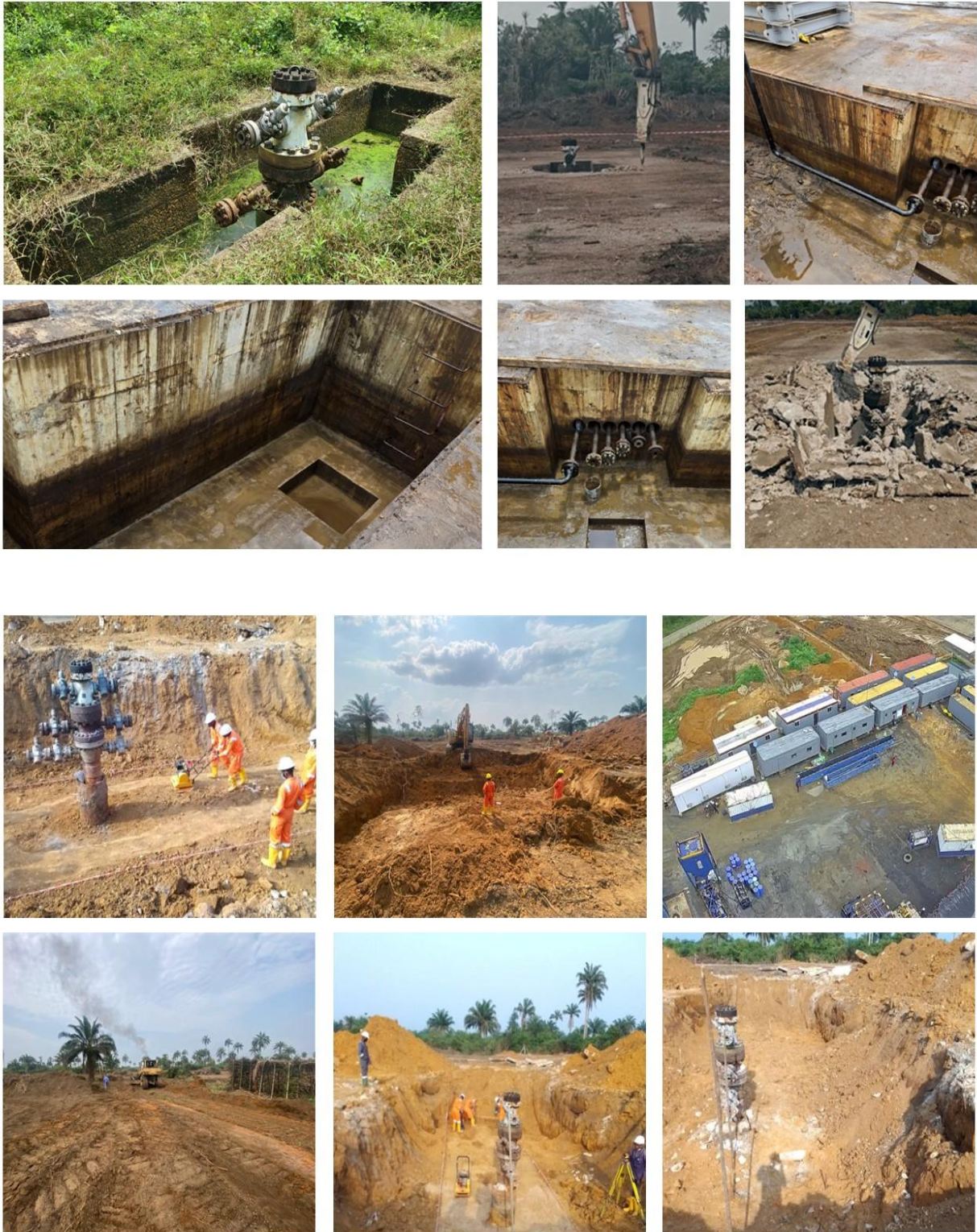
Axis delivered comprehensive well engineering for the Emohua field, including re-entry planning of a legacy well drilled in 1979, new well drilling (Emohua-2), dual-string completion design with Long String (deep reservoir) and Short String (shallower zone), formation-specific gravel pack sand control, and full production system readiness.

### Key Challenges & Axis Solutions

| Challenge                              | Axis Solution                                                           |
|----------------------------------------|-------------------------------------------------------------------------|
| Legacy well uncertainty (1979 vintage) | Comprehensive re-entry assessment with contingency planning             |
| Sand production risk                   | Formation-tailored gravel pack design with QA/QC                        |
| Completion integrity across dual zones | Engineered dual-string completion architecture                          |
| Multi-vendor coordination complexity   | Structured DWOP-driven execution with tight SLB/Weatherford integration |

### Results

- ✔ Successful well delivery across both Long String and Short String zones
- ✔ Multi-zone reservoir access achieved with controlled sand production
- ✔ Full well lifecycle execution capability demonstrated in Niger Delta conditions



## Location Redesign and Preparation





## Rig Inspections & Selection

## Case Study 2: Kuri Field (PPL 222)

### Swamp Asset — Heavy Oil Production Activation

**Project Overview:** Kuri-2 well required re-entry and production activation in a challenging swamp environment. The heavy oil reservoir (9.8 ppg NaCl completion fluid) presented unique flow initiation challenges requiring an engineered unloading approach.

### Axis Unloading Strategy

Our team designed and executed a structured five-step well activation program:

| Step | Phase                   | Description                                                                          |
|------|-------------------------|--------------------------------------------------------------------------------------|
| 1    | Fluid Displacement      | Controlled displacement to avoid formation damage while clearing completion fluid    |
| 2    | Nitrogen Lifting        | N <sub>2</sub> kick-off to initiate flow in heavy oil column                         |
| 3    | Pressure Drawdown       | Gradual drawdown to establish stable inflow without sand failure                     |
| 4    | Flowback Monitoring     | Real-time monitoring of rates, pressures, and fluid properties                       |
| 5    | Well Test Stabilization | Sustained flow period to confirm reservoir deliverability and fluid characterization |

### Results

- ✓ Safe unloading approach defined and executed with structured 15-day cleanup plan
- ✓ Controlled transition to production phase with heavy oil handling capability demonstrated
- ✓ Integrated CTU + Nitrogen services (Weafri) with marine and security coordination (Tantita)

“Axis has unlocked difficult wells and converted them into producing assets.”





## Extended Well Test (Swamp + Land Assets)

## Case Study 3: Field Development Planning (FDP)

### Extending Our Reach — Nigeria & Republic of Congo

Beyond well-level execution, Axis provides full-scope Field Development Planning (FDP) services that translate subsurface data into bankable, executable development strategies. Our FDP capability is particularly valuable for marginal field awardees and junior E&P companies seeking to move from exploration data to investment-ready development plans. The following engagements demonstrate our FDP competence across two distinct operating environments.

#### A. Petrokinetics Energy — Nigeria

**Client:** Petrokinetics Energy Limited

**Scope:** Full Field Development Plan for a marginal field asset in Nigeria's Niger Delta

#### Project Background

Petrokinetics Energy engaged Axis Well Delivery Systems to develop a comprehensive Field Development Plan for their marginal field asset. As a marginal field awardee, Petrokinetics required a technically robust yet commercially bankable FDP that would satisfy NUPRC regulatory requirements, attract investment partners, and provide a clear roadmap from current state to sustained production.

#### Axis Scope of Work

Our team conducted a thorough subsurface evaluation, integrating available well logs, seismic data, and offset well performance to characterize the reservoir and define producible reserves. We developed multiple development scenarios — evaluating well count, trajectory options, completion strategies, and artificial lift requirements — to identify the optimal path to first oil with the lowest capital exposure.

The FDP deliverables included reservoir characterization and volumetric estimates, well location and trajectory planning with anti-collision analysis, casing and completion design tailored to Niger Delta sand-prone formations, surface facility concept selection and sizing, production forecasting with decline curve analysis, drilling and completion cost estimates with detailed AFE build-up, project execution schedule with critical path milestones, and a comprehensive risk register with mitigation strategies.

## Key Value Delivered

- ✓ NUPRC-compliant FDP document delivered on schedule, supporting the client's regulatory submission
- ✓ Multiple development scenarios provided with clear economic comparisons (NPV, IRR, payback)
- ✓ Phased development approach recommended to reduce upfront capital and de-risk the investment
- ✓ Sand control strategy engineered from offset well data, reducing completion risk for the first well
- ✓ FDP package structured for investor presentation, accelerating the client's fundraising timeline

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“Axis transformed Petrokinetics' subsurface data into an investment-ready development roadmap.”

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**B. Olive Energy — Republic of Congo (Brazzaville)**

**Client:** Olive Energy Congo SA

**Scope:** Field Development Plan and Well Engineering for onshore assets in the Republic of Congo

**Project Background**

Olive Energy is a junior integrated oil and gas company headquartered in Pointe-Noire, Republic of Congo, focused on de-risking exploration blocks and increasing production from brownfield and marginal assets. Operating the Tilapia permit and pursuing expansion through the Marine III license, Olive Energy required a technically rigorous FDP that addressed the unique geological and operational challenges of the Congolese onshore environment.

Olive Energy engaged Axis Well Delivery Systems to provide Field Development Planning and well engineering services, leveraging our proven expertise in brownfield revitalization and marginal field development. This engagement marked Axis’s first cross-border FDP assignment, demonstrating our capability to deliver beyond Nigeria into the wider African energy market.

**Axis Scope of Work**

Working closely with Olive Energy’s subsurface and operations teams, Axis delivered an integrated FDP covering reservoir review and resource assessment using available well and seismic data, development well planning including trajectory design, casing programmes, and drilling hazard assessment for the Congo basin geology.

Our team also provided completion and sand management strategies adapted to the formation characteristics, artificial lift evaluation for the expected fluid properties and well productivity, production facility concept and sizing aligned with Olive Energy’s modular development philosophy, detailed cost estimates and economic modelling across multiple development scenarios, and an execution schedule incorporating local logistics, regulatory, and community engagement timelines specific to the Republic of Congo.

**Key Challenges Addressed**

| Challenge | Axis Approach |
|-----------|---------------|
|-----------|---------------|

|                                                                      |                                                                                                                   |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Limited well data in a frontier-type onshore basin                   | <b>Integrated offset analogue study from Niger Delta and Congo basin fields to fill data gaps</b>                 |
| Adapting Niger Delta engineering standards to Congolese geology      | <b>Customized well designs accounting for local pore pressure regimes and formation properties</b>                |
| Aligning FDP with SNPC regulatory framework                          | <b>FDP structured to meet Republic of Congo petroleum regulations and SNPC partnership requirements</b>           |
| Modular, phased development to match junior operator capital profile | <b>Scrum-style phased approach with 6-month sprint milestones, matching Olive Energy's development philosophy</b> |

## Results & Impact

- ✓ Comprehensive FDP delivered covering well engineering, production forecasting, and facility design
- ✓ Development plan aligned with Olive Energy's partnership model with SNPC (Société Nationale des Pétroles du Congo)
- ✓ Phased execution strategy designed to match capital availability and reduce investment risk
- ✓ Cross-border capability demonstrated — proving Axis can deliver beyond Nigeria into wider African markets
- ✓ Well engineering designs adapted for Congo basin conditions while maintaining international standards

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“Our engagement with Olive Energy Congo demonstrates Axis's readiness to serve as a pan-African well delivery partner — bringing Niger Delta expertise to new frontiers.”

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## Case Study 4: Rig Inspections & Selection

### Ensuring Operational Readiness Before Mobilization

Rig selection is one of the most critical decisions in any well delivery programme. A poorly maintained or ill-suited rig can cause catastrophic delays, safety incidents, and cost overruns that derail an entire campaign. Axis Well Delivery Systems conducts rigorous, independent rig inspections to verify mechanical integrity, equipment suitability, and regulatory compliance before any mobilization commitment.

Our rig inspection methodology covers structural and mechanical assessment, BOP and well control equipment testing and certification review, power systems and electrical integrity, mud circulating and solids control systems, mast/derrick and hoisting equipment, HSE systems including fire suppression, gas detection, and emergency response readiness, crew competence verification, and regulatory documentation audit. The following five inspections demonstrate our capability across international and domestic rig markets.

| # | Rig Name             | Type / Spec                 | Location                           | Client                      | Outcome                                            |
|---|----------------------|-----------------------------|------------------------------------|-----------------------------|----------------------------------------------------|
| 1 | Jewel Quest          | Swamp Barge<br>3000HP / 10K | New Iberia,<br>Louisiana,<br>USA   | Platform<br>Petroleum Ltd   | Inspected for swamp<br>operations capability       |
| 2 | Acme Rig 7           | Land Rig<br>2000HP          | Conroe,<br>Texas, USA              | EOP Energy<br>Resources Ltd | Inspected for Nigeria<br>land drilling<br>campaign |
| 3 | Melcurt Rig 1        | Land Rig                    | Okpaka,<br>Delta State,<br>Nigeria | EOP Energy<br>Resources Ltd | Selected — used for<br>Emohua-1 re-entry           |
| 4 | MF101<br>(Matpatson) | Land Rig                    | Nigeria                            | EOP Energy<br>Resources Ltd | Selected — used for<br>Emohua-2 & 2 ST             |
| 5 | Acme Rig 5           | Land Rig                    | Maiduguri,<br>Nigeria              | EOP Energy<br>Resources Ltd | Inspected for<br>Northeast operations              |

## 1. Jewel Quest — Swamp Barge Rig (Platform Petroleum Ltd)

**Rig:** Jewel Quest — 3,000 HP, 10,000 psi rated swamp barge drilling rig

**Location:** New Iberia, Louisiana, USA

**Client:** Platform Petroleum Ltd

Platform Petroleum Ltd required a high-specification swamp barge rig capable of operating in the challenging Niger Delta swamp terrain. Axis was engaged to conduct a comprehensive pre-purchase inspection of the Jewel Quest in New Iberia, Louisiana. Our inspection team assessed the structural integrity of the barge hull and deck, the condition and certification status of the 3,000 HP drawworks, mud pumps, and rotary system, BOP stack and well control equipment rated to 10,000 psi working pressure, power generation systems including SCR/VFD drives, and the overall suitability of the rig package for Nigerian swamp operations including draft requirements, anchor systems, and environmental compliance.

The inspection provided Platform Petroleum with a detailed condition report, a punch-list of remediation items required prior to mobilization, estimated refurbishment costs, and a clear recommendation on the rig's fitness for purpose in their planned Niger Delta swamp drilling campaign.

## 2. Acme Rig 7 — Land Rig (EOP Energy Resources Ltd)

**Rig:** Acme Rig 7 — 2,000 HP land drilling rig

**Location:** Conroe, Texas, USA

**Client:** EOP Energy Resources Ltd

EOP Energy Resources engaged Axis to inspect the Acme Rig 7 in Conroe, Texas as part of their rig sourcing programme for Nigerian land operations. Our team conducted a full mechanical and structural assessment covering the mast, substructure, drawworks, mud system, and power packages. We evaluated the rig's compatibility with the planned well depth and casing programme, assessed the BOP and well control equipment certification, and reviewed the rig's maintenance history and operational records. The inspection report enabled EOP to make an informed acquisition decision with a clear understanding of any required upgrades before deployment to Nigeria.

## 3. Melcurt Rig 1 — Land Rig (EOP Energy Resources Ltd)

**Rig:** Melcurt Rig 1 — Land drilling rig

**Location:** Okpaka, Delta State, Nigeria

**Client:** EOP Energy Resources Ltd

Axis inspected the Melcurt Rig 1 at its stacked location in Okpaka, Delta State. This was a particularly critical inspection as the rig was being evaluated for the Emohua-1 legacy well re-entry — a complex operation requiring precise well control capability and reliable equipment in a brownfield environment. Our assessment focused on the rig's readiness for immediate mobilization, the condition of key drilling equipment after an extended stacking period, BOP and well control certification status, and the overall suitability for the re-entry operation's specific requirements including the well's 1979 vintage casing and wellhead configuration.

**Outcome:** Melcurt Rig 1 was selected and subsequently used for the successful Emohua-1 re-entry operation, validating our inspection methodology and rig selection process.

#### 4. MF101 (Matpatson) Land Rig (EOP Energy Resources Ltd)

**Rig:** MF101 (Matpatson) Land drilling rig

**Location:** NPA Port Harcourt, Nigeria

**Client:** EOP Energy Resources Ltd

Following the successful Emohua-1 re-entry, EOP required a rig for the new-build Emohua-2 well and its subsequent sidetrack (Emohua-2 ST). Axis inspected the MF101 (Matpatson) rig, evaluating its capability for both the primary well and the planned sidetrack operation. The inspection scope covered drilling equipment capacity for the planned well depths, mud system adequacy for the anticipated formation pressures, BOP and well control equipment condition and certification, and the rig's track record and crew competence for sidetrack operations requiring precise directional control.

**Outcome:** MF101 was selected and successfully drilled both Emohua-2 and Emohua-2 ST, completing the Emohua field development campaign.

#### 5. Acme Rig 5 — Land Rig (EOP Energy Resources Ltd)

**Rig:** Acme Rig 5 — Land drilling rig

**Location:** Maiduguri, Nigeria

**Client:** EOP Energy Resources Ltd



Axis conducted an inspection of the Acme Rig 5 at its location in Maiduguri, in Nigeria's northeast. This inspection presented unique logistical challenges given the remote location and security considerations in the region. Our team assessed the rig's mechanical condition, equipment certification status, and operational readiness, providing EOP with a comprehensive evaluation of the rig's suitability for their planned drilling programme.

### **Rig Inspection Capability Summary**

Across these five inspections spanning the United States (Louisiana and Texas) and Nigeria (Delta State, Maiduguri), Axis demonstrated the breadth and depth of our rig assessment capability. Two of the five inspected rigs — Melcurt Rig 1 and MF101 (Matpatson) were ultimately selected and used to successfully execute the Emohua field drilling campaign, directly linking our rig inspection quality to operational success.

- ✓ Five rigs inspected across four locations in two countries (USA and Nigeria)
- ✓ Both swamp barge and land rig configurations assessed
- ✓ Two inspected rigs selected and deployed — delivering successful wells at Emohua
- ✓ Inspection methodology validated by zero rig-related NPT on subsequent operations

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“Axis doesn't just select rigs — we verify them. Our inspection rigor ensures operational readiness before a single dollar is committed to mobilization”.

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## Case Study 5: Ede E&P Ltd — Ede-4 Well Engineering

### Offshore Marginal Field — Multi-Zone Development Well

**Client:** Ede E&P Ltd (Marginal Field Operator)

**Asset:** Ede Field — Offshore Marginal Field

**Scope:** Full well engineering and delivery planning for Ede-4, a high-integrity multi-zone development well

**Status:** Engineering complete — approaching execution phase

### Project Background

Ede E&P Ltd, a marginal field operator preparing to move into the execution phase, engaged Axis Well Delivery Systems to provide comprehensive well engineering for their Ede-4 development well. Ede-4 is a technically demanding multi-zone well designed to penetrate and complete three reservoir zones (Upper, Middle, and Deep A6), with the deep zone presenting significant overpressure challenges requiring a robust barrier philosophy.

The well's complexity stems from the requirement for dual-string completion across zones with markedly different pressure regimes, the need for selective zonal production with sand control, and the critical importance of maintaining well integrity across the full lifecycle from drilling through completion and into sustained production.

### Axis Scope of Work

Axis delivered a comprehensive well engineering package covering every aspect of the Ede-4 well design and execution planning:

| Engineering Discipline            | Deliverables                                                                                                                                                                                         |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Well Architecture & Casing Design | Conductor/surface, 9 $\frac{5}{8}$ " 47ppf N80 VAM TOP intermediate, 7" VAM TOP (HT) production liner. Full hole section sizing for 12 $\frac{1}{4}$ " and 8 $\frac{1}{2}$ " sections.               |
| Casing & Tubing Engineering       | WellCAT-philosophy load case design covering running, cementing, production, and worst-case scenarios (gas kick, trapped annulus). Burst, collapse, tension, and triaxial checks across all strings. |
| Torque & Drag Analysis            | Critical analysis for rotating liner while cementing: 5" S-135 string, friction factor sensitivity (0.30/0.40/0.50), 80% surface torque cap policy. VAM TOP HT connections specified.                |

|                                   |                                                                                                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cementing Programme</b>        | 15.8 ppg slurry design, ~217 bbl volume, 6 bpm pump rate. Rotation during cementing for displacement efficiency. Spacer optimization for mud removal.                                                |
| <b>Completion Strategy</b>        | Dual-string architecture: Long String to deep zone (A6), Short String to upper zones. 2 <sup>7</sup> / <sub>8</sub> " L-80 tubing, production packers, SCSSV, seal assemblies, tubing hanger system. |
| <b>Barrier Philosophy</b>         | Comprehensive barrier design for overpressured A6 zone (≥300 psi overbalance, 15.2 ppg CaBr <sub>2</sub> brine). Two barrier modes defined with full verification protocols.                         |
| <b>Trajectory &amp; Targeting</b> | Directional well with build section, kickoff at ~7,700 ft MD (~6,399 ft TVD). D-series sand landing, fault positioning, and TVD/MD mapping.                                                          |
| <b>Risk Register</b>              | Five major risks identified and mitigated: stuck pipe, poor cement job, pressure control failure, torque limit exceedance, and annular pressure build-up.                                            |

## Technical Highlights

The Ede-4 well engineering presented several technically demanding challenges that showcase Axis's depth of capability:

**Overpressured Deep Zone (A6):** The A6 reservoir's overpressure governs the entire burst envelope and requires heavy  $\text{CaBr}_2$  completion brine at 15.2 ppg with a minimum 300 psi overbalance. Axis designed a comprehensive barrier philosophy with two defined barrier modes — Mode 1 for A6 isolation (sealed below FIV with heavy brine in place) and Mode 2 for upper zone perforation while A6 remains isolated. Every barrier element was specified with verification protocols including pressure tests, leak-off validation, SCSSV functional tests, and annulus monitoring.

**Liner Rotation While Cementing:** Achieving a quality cement job across the production interval requires rotating the 7" liner during cement displacement — a technically demanding operation. Axis performed detailed torque and drag analysis at multiple friction factors, confirmed feasibility with VAM TOP HT connections, and established an 80% surface torque cap policy with real-time monitoring requirements. A reciprocation fallback was built into the contingency plan.

**Dual-Zone Completion Architecture:** The completion design enables selective production from three reservoir zones using a dual-string configuration. The long string accesses the deep A6 zone while the short string targets the upper zones, with production packers and seal assemblies providing zonal isolation. Material compatibility was verified for  $\text{CaBr}_2$  brine exposure on L-80 tubing for the anticipated operational duration.

## Key Value Delivered

- ✓ Technically sound, risk-aware well design validated through comprehensive load case analysis (WellCAT philosophy)
- ✓ Robust barrier philosophy with two defined modes for safe multi-zone operations in overpressured environment
- ✓ Torque & drag analysis confirming liner rotation feasibility — critical for cement quality
- ✓ Dual-zone production capability designed to maximize reservoir recovery across three zones
- ✓ Complete risk register with five major risks identified and mitigated before execution

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- ✔ Full execution-ready engineering package: well programme, barrier diagrams, cementing programme, and DWOP-ready documentation
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“Ede-4 demonstrates Axis’s ability to deliver execution-ready well engineering for complex, multi-zone offshore development wells — taking marginal field operators from concept to the threshold of first oil.”

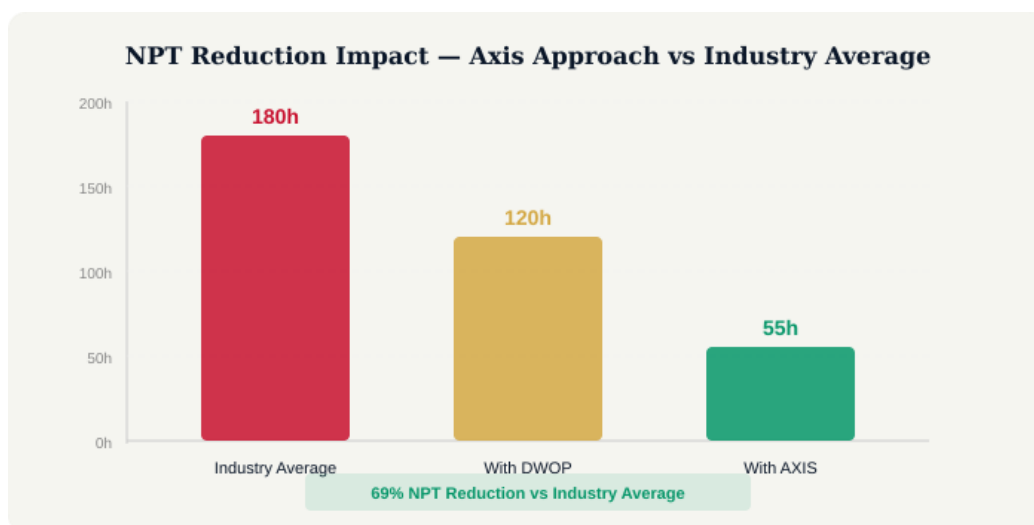
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# Cost & NPT Management Framework

**“We eliminate waste to accelerate your ROI.”**

Axis employs a rigorous cost and non-productive time (NPT) management framework designed to deliver wells on budget and ahead of schedule. Our approach combines proactive planning, real-time decision authority, and vendor accountability to minimize waste and maximize operational efficiency.



## NPT Reduction Strategy

- Pre-job DWOP alignment ensures all parties are synchronized before operations begin
- Real-time decision authority at the rig site eliminates delays waiting for remote approvals
- Vendor accountability structure with KPIs and performance incentives

## Stuck Pipe Contingency

- Pre-defined fishing strategy with decision trees documented in every well program
- Torque & drag modeling to predict and prevent stuck pipe scenarios
- Contingency tools on standby — fishing jars, bumper subs, and overshots pre-mobilized

## Sand Control Assurance

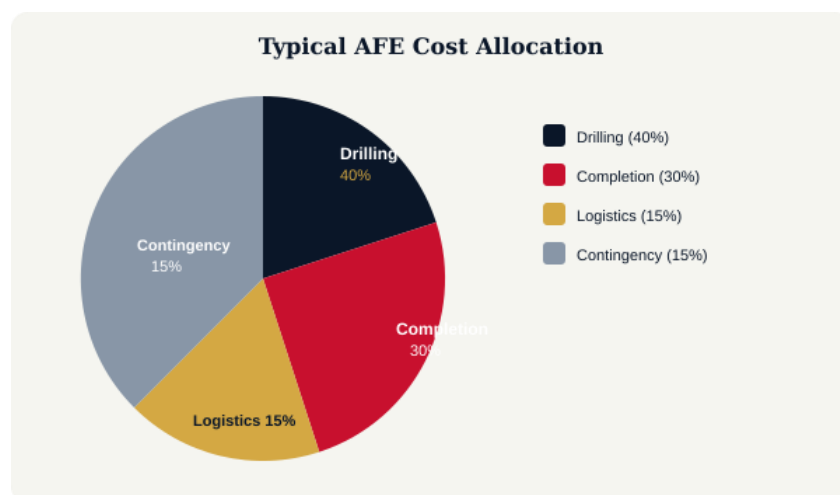
- Formation-specific gravel pack design based on sieve analysis and core data
- QA/QC on installation with real-time pump rate and pressure monitoring

 Post-completion monitoring to verify screen and gravel pack integrity

## AFE Tracking & Commercial Value

Our disciplined AFE management consistently delivers wells below budget. The following illustrative framework demonstrates our approach to cost tracking and variance management:

| Cost Element                  | AFE Budget (USD)   | Actual (USD)       | Variance      |
|-------------------------------|--------------------|--------------------|---------------|
| <b>Rig &amp; Spread</b>       | \$2,800,000        | \$2,650,000        | <b>-5.4%</b>  |
| <b>Drilling Services</b>      | \$1,500,000        | \$1,420,000        | <b>-5.3%</b>  |
| <b>Completion Equipment</b>   | \$1,200,000        | \$1,180,000        | <b>-1.7%</b>  |
| <b>Logistics &amp; Marine</b> | \$600,000          | \$580,000          | <b>-3.3%</b>  |
| <b>Contingency (10%)</b>      | \$610,000          | \$120,000          | <b>-80.3%</b> |
| <b>TOTAL</b>                  | <b>\$6,710,000</b> | <b>\$5,950,000</b> | <b>-11.3%</b> |



Typical Savings: 10–15% Below AFE

# Organizational Structure

**Lean = Lower Cost = Competitive Advantage.** Our purpose-built team structure minimizes overhead while maximizing operational capability. Every role is designed to add direct value to well delivery outcomes.



Our lean organizational model enables rapid decision-making, direct accountability, and cost efficiency that larger organizations cannot match. This structure delivers IOC-grade results without the massive overhead — a critical advantage for marginal field operators and indigenous E&P companies.

## Target Market Focus

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### **Marginal Field Awardees**

**From Award to First Oil.** Marginal field awardees need a technical partner who can take them from license award through well design, drilling, completion, and into sustained production. Axis provides end-to-end well delivery so you can focus on asset strategy and commercial optimization while we handle the technical execution.

### **Indigenous E&P Companies**

**Your Outsourced Wells Department.** Many indigenous operators do not maintain a permanent wells team. Axis acts as your integrated wells department — handling planning, drilling, completions, and intervention with full accountability. This model eliminates the cost of maintaining a large technical staff while ensuring access to experienced professionals when you need them.

### **IOC Divestment Assets**

**Extending Asset Economic Life.** As international oil companies divest mature Nigerian assets, the new operators need technical partners who understand brownfield challenges. Axis brings deep expertise in well revitalization, workover programs, and production optimization to maximize recovery from aging wells and extend the economic life of legacy fields.

## Contact Us

Partner with Axis to transform your subsurface potential into producing assets.

|                                                                                                |                                                                                    |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
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Axis Well Delivery Systems Ltd — Engineered Right, Creating Value

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Transforming subsurface potential into producing assets — safely, efficiently, and profitably.